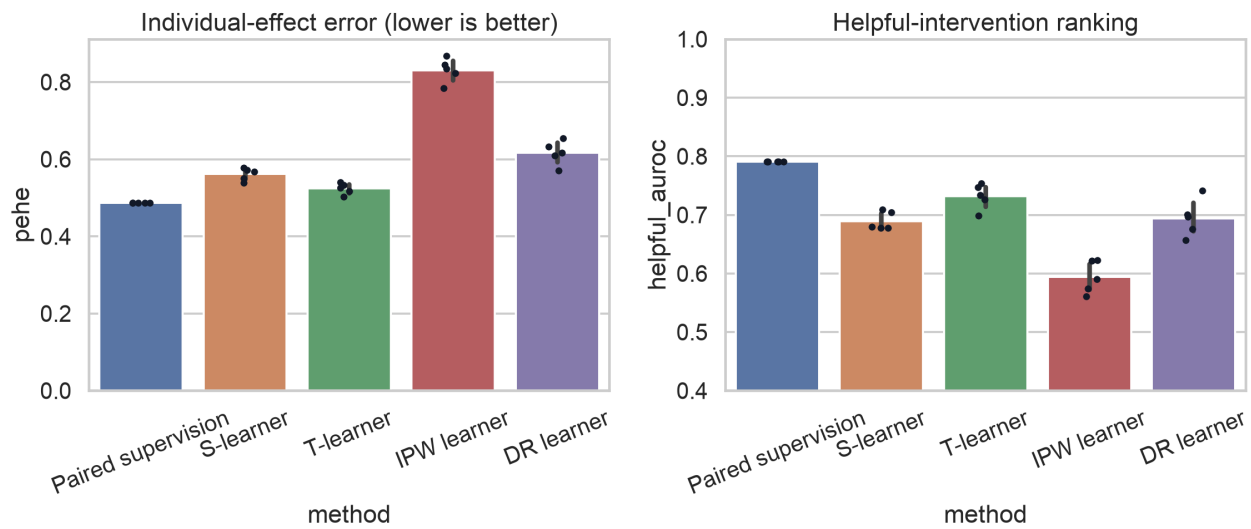


Learning When to Ask from One Observed Future

A hidden-counterfactual benchmark for selective robot assistance



Exact treatment-effect error and helpful-intervention ranking. Dots are factual logging seeds; paired supervision is privileged.

0.524	0.732	240
T-learner PEHE	helpful AUROC	one-outcome episodes

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Simulation-only. Free and reproducible on Apple Silicon or CPU. No CUDA, cloud compute, paid API, physical robot, or human-subject claim.

Abstract

An autonomous robot can observe the outcome of continuing alone or requesting assistance, but not both. Prior InterveneSim experiments used exact simulator forks as paired training labels, which cannot be obtained from a physical deployment. InterveneSim-CF moves those forks behind an evaluation boundary. Each training episode exposes one candidate context, one logged treatment, one realized terminal outcome, and the known logging propensity. We compare S-, T-, inverse-propensity, and cross-fitted doubly robust effect learners under randomized, uncertainty-selective, and online adaptive logging. Exact hidden forks then measure individual treatment-effect error and policy value on unseen policy seeds. With 240 factual episodes, the T-learner reaches 0.524 precision in estimation of heterogeneous effect (PEHE) and 0.732 AUROC for helpful interventions, versus 0.614 and 0.706 for failure risk. Its equal-budget task-success gain over risk is nevertheless only 0.4 points at 25% intervention and 2.0 points at 50%, and it does not consistently outperform a reversibility proxy. Doubly robust learning sharply improves over uncorrected inverse propensity weighting under selective logging, but its training-calibrated 25% threshold requests help on roughly half of unseen episodes. Leave-one-task-out, leave-one-disturbance-out, cost, sample-efficiency, and assistance-latency audits retain the negative cases. The result is a deployment-realistic benchmark and a caution: estimating causal benefit better does not by itself solve selective robot assistance.

1. Motivation

Interactive imitation learning asks an expert to label or control states encountered by a learner. Query policies commonly use novelty, uncertainty, or predicted failure. Those signals are useful but do not identify the causal question: will this particular assistance treatment change the eventual outcome?

PAINT learns to request help before an agent enters an irreversible state [1]. ThriftyDagger combines novelty and risk under a query budget [2]. HG-Dagger lets a human supervisor decide when to take control [3]. InterveneSim studies an adjacent but distinct estimand: signed outcome value for a particular assistant. A state may be dangerous after the recovery window has closed, or it may look unfamiliar while the policy remains capable. Assistance can also be ineffective or harmful.

InterveneSim-Value v0.3 measured that quantity by restoring an exact MuJoCo snapshot and replaying autonomous and assisted futures. Its value gate outperformed risk when both futures were provided as labels. A physical trajectory cannot be rewound to reveal the unchosen future. This work asks what survives after removing that privileged supervision.

2. Hidden-counterfactual benchmark

The source archive contains 1,505 training candidates from 240 episodes generated by policy seeds 27, 127, and 227. The evaluation archive contains 1,517 candidates from 240 episodes generated by unseen policy seeds 327 and 427. Tasks manipulate can, milk, bread, and cereal objects under nominal operation, object shift, action noise, action delay, and gripper slip. Candidate contexts concatenate the current task-conditioned observation, its causal one-step change, and the autonomous action.

Two schemas enforce the information boundary. `CounterfactualData` contains both potential outcomes and is accepted only by the logger and evaluator. `LoggedBanditData` contains context, selected treatment, realized outcome, propensity, episode identity, task, disturbance, and step. Its serializer cannot write either potential outcome, helpfulness, or signed value. Every frozen log is scanned by a machine-readable leakage audit.

Each source episode contributes one uniformly sampled candidate. This avoids treating seven candidate states from one autonomous trajectory as independent terminal interventions. The logger selects immediate scripted assistance or autonomous continuation and reveals only the chosen arm. Five logging seeds vary both candidate time and treatment realization.

3. Logging policies and estimators

The randomized logger assigns assistance with probability 0.5. The uncertainty-selective logger uses policy-ensemble action variance and clips propensities to $[0.1, 0.9]$. The adaptive-value logger begins with a randomized batch, refits an S-learner every 60 episodes using only past factual outcomes, and applies an epsilon-soft treatment policy based on predicted value. Every propensity remains known.

The S-learner fits one outcome model with treatment as an input. The T-learner fits autonomous and assisted outcome models separately with inverse-propensity weights. The IPW learner regresses the standard inverse-propensity pseudo-outcome. The doubly robust learner cross-fits both nuisance outcome models by episode, constructs an augmented inverse-propensity pseudo-outcome, and fits an effect model. Small MLP heads run on CPU or Apple Metal.

The reversibility proxy learns autonomous failure from factual control outcomes and assumes that help repairs the failure. It approximates the risk assumption needed for comparison but is not a reproduction of PAINT's online reinforcement-learning procedure. Failure risk and ensemble uncertainty from the prior release are also retained. A paired-label regressor sees exact signed effects and serves only as a privileged supervision comparator.

4. Evaluation

Hidden exact effects are assisted success minus autonomous success. Primary estimator metrics are PEHE, average-effect error, helpful-intervention AUROC and average precision, harmful ranking, and sign accuracy. Primary policy comparisons score-rank episodes and allocate the exact same 10%, 25%, 50%, or 75% intervention budget without reading outcomes. This isolates ranking quality from score calibration.

A second deployment audit freezes thresholds using unlabeled training contexts and reports the realized unseen-policy intervention rate. Cost-sensitive policies maximize success minus an explicit intervention cost. Learning curves use 60, 120, 180, and 240 factual episodes. Open-world audits remove each task and disturbance from training in turn. A latency audit reuses exact later forks along the autonomous path to delay the same assistant by one or two candidate intervals.

Logging seed is the unit of method uncertainty. Candidate states are not treated as independent physical trials. Bootstrap intervals are descriptive, and all hypotheses remain exploratory because the v0.3 archive and pilots were already available.

5. Results

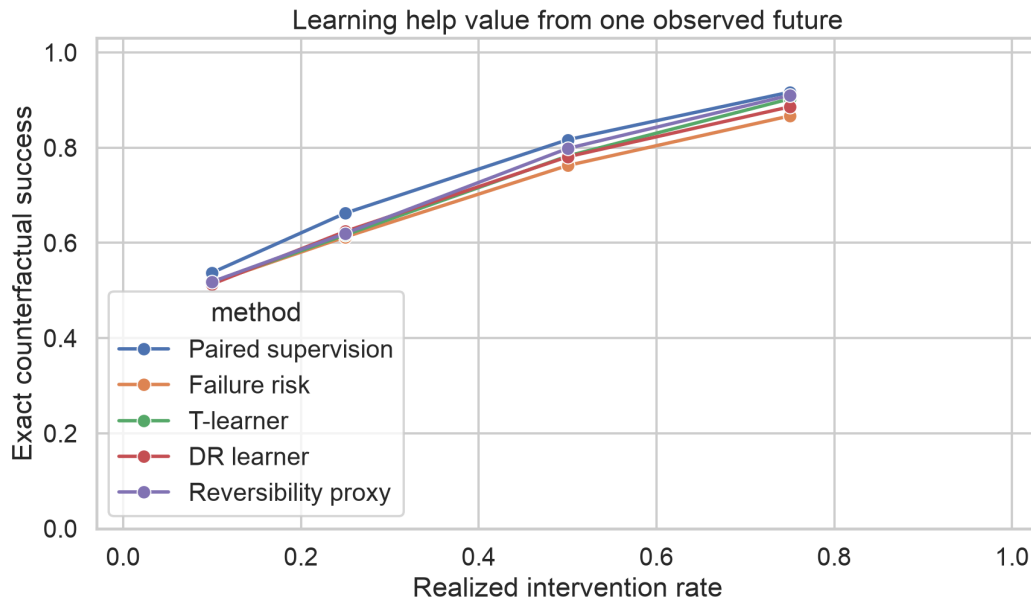


Figure 2. Equal-budget success on unseen policy seeds. Every method receives the exact same episode-level intervention budget.

Method	PEHE	AUROC	Success @25%	Success @50%
Paired supervision	0.488	0.791	66.2%	81.7%
T-learner	0.524	0.732	61.7%	78.2%
DR learner	0.617	0.694	62.4%	78.1%
Reversibility	0.550	0.725	62.0%	79.8%
Failure risk	0.614	0.706	61.3%	76.2%

Under randomized logging, the T-learner obtains mean PEHE 0.524, compared with 0.550 for the reversibility proxy, 0.614 for failure risk, and 0.488 for privileged paired-label supervision. Its helpful AUROC is 0.732, versus 0.725, 0.706, and 0.791 respectively. The T-learner's average-effect error is 0.054, while failure risk—an intentionally non-causal score—has error 0.221.

At an exactly matched 25% intervention rate, T-learning reaches 61.7% mean success, failure risk 61.3%, reversibility 62.0%, and paired-label supervision 66.2%. At 50%, the corresponding rates are 78.2%, 76.2%, 79.8%, and 81.7%. The single-world learner therefore recovers useful effect signal but does not reproduce v0.3's large policy advantage.

The cross-fitted doubly robust learner is substantially more stable than raw IPW under selective logging: PEHE is 0.608 versus 0.847. This supports propensity correction but does not make DR the best learner in this small sample; T-learning remains better at 0.523 PEHE. Negative comparisons are retained.

Calibration is a separate failure. Thresholds chosen to target 25% intervention on training contexts cause the DR learner to request help on about half of unseen-policy episodes. Reporting only its resulting success would create an invalid comparison with risk. The exact-budget result removes that artifact, while the deployment table preserves it as evidence that ranking transfer does not guarantee threshold transfer.

6. Distribution and assistant shift

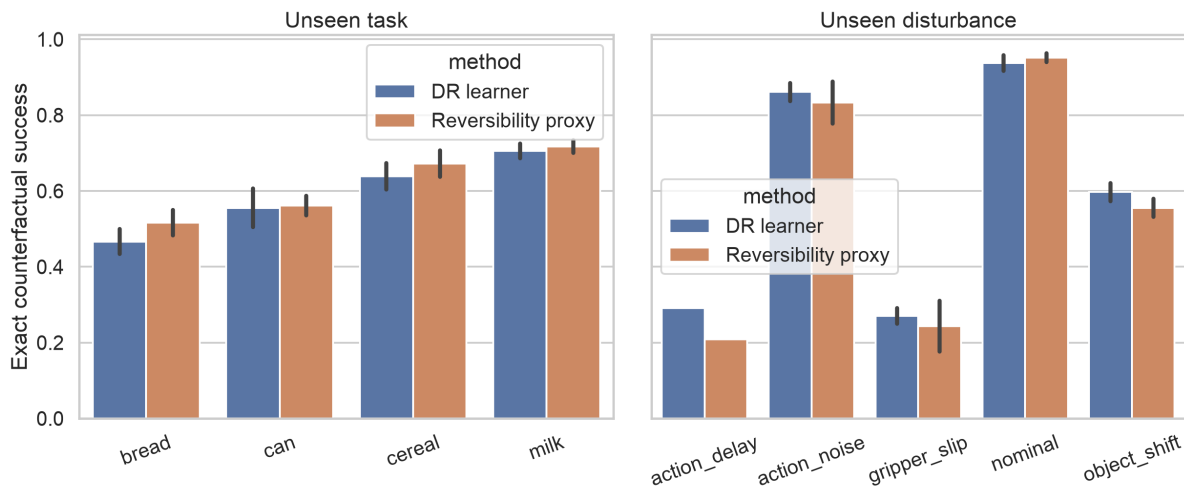


Figure 3. Leave-one-task-out and leave-one-disturbance-out success near 25% intervention.

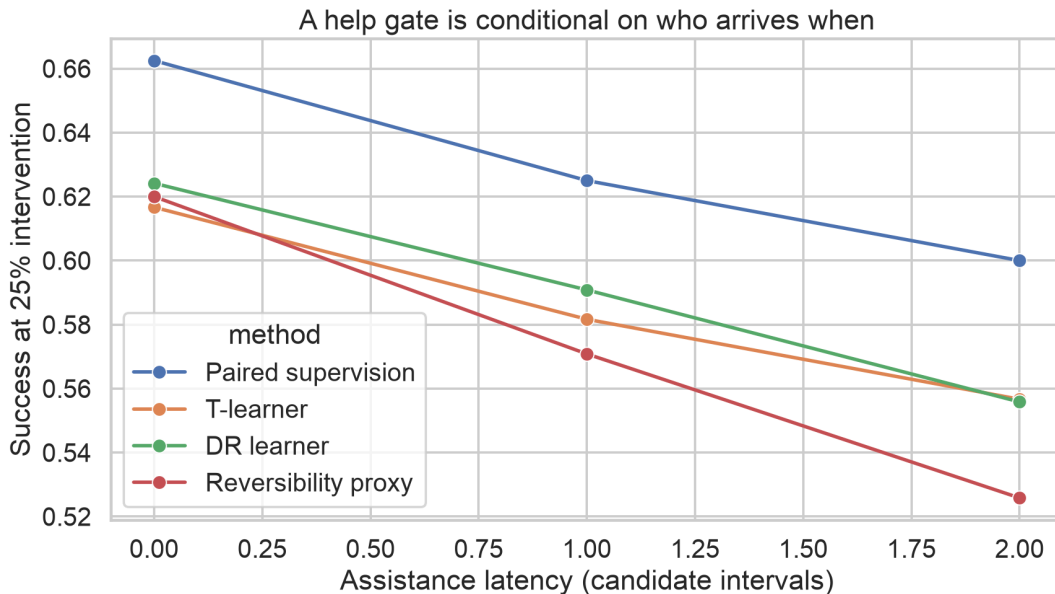


Figure 4. Immediate and delayed assistance under the same learned gate and exact autonomous trajectories.

Leave-one-group-out results vary widely. Near a 25% budget, mean success across held-out tasks is approximately 59% for DR and 62% for the reversibility proxy; across held-out disturbances it is approximately 59% and 56%. Aggregating these heterogeneous groups can conceal severe failures, so the repository publishes every group and logging seed.

The latency audit changes the treatment while holding the request state fixed. Immediate help, one-candidate-delayed help, and two-candidate-delayed help use exact branches from progressively later states on the same autonomous trajectory. Performance degradation demonstrates that intervention value is conditional on assistant timing. These profiles do not stand in for human operators with different skills.

Learning curves show that additional factual episodes generally reduce PEHE and improve helpful ranking, but individual seeds are not monotone. The online adaptive logger is an audit of selective data collection, not an online improvement to the robot control policy.

7. Threats to validity

Exact forks are exact only for the implemented simulator, scripted assistant, disturbance process, and decision grid. The logger is simulated from paired archives, although its output faithfully hides the unchosen arm. Reusing source episodes across logging seeds measures missing-outcome sensitivity rather than independent physical replication. New environment episodes are required for confirmatory scaling.

The reversibility baseline is a controlled proxy and must not be presented as a faithful PAINT reproduction. The assistant has privileged geometry and bypasses synthetic actuator corruption. Delayed profiles follow later candidates from the original autonomous trajectory; they do not model communication, human reaction, or partial takeover. No result establishes real-robot safety, human intervention quality, or sim-to-real transfer.

8. Conclusion

InterveneSim-CF converts deterministic simulator rewinds from unrealistic training labels into a hidden counterfactual evaluator. Single-world effect learners recover meaningful causal signal, and propensity-aware methods avoid some selective-logging error. Yet equal-budget policy gains are small, reversibility remains competitive, calibration shifts sharply, and latency changes the treatment. Those limitations are the main result: learning when to ask is not solved by replacing risk with a causal score unless data scale, calibration, distribution shift, and assistant identity are handled together.

References

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